Fundamentals of Pharmaceutical Chemistry

What is pharmaceutical chemistry ?

Pharmaceutical chemistry is the study of drugs and it involves drug development, drug discovery, delivery, absorption, metabolism and more.

Pharmaceutical Raw Materials?

Pharmaceutical raw materials are substances , chemicals, extracts, substrate or element that are used for manufacturing different types of drugs. Ex: Penicillin, Ampicillin

Raw material sources

- Natural
- Synthetic

There are three major types of pharmaceutical raw materials

- Active pharmaceutical ingredients (API)
- Excipients
- Raw material packaging

Drying

- Drying is defined as removal of liquid from the materials by application of heat and is accompanied by the transfer of liquid from a surface to an unsaturated vapor state.
- Drying applies for the removal of a small amount of liquids from the matters. Drying and evaporation are distinguished by the relative quantity of liquid removed from the solid.

Purpose of drying:

1. Decrease the bacterial growth in solvent
2. For preparation of granules
3. Reduce the bulk weight
4. Dried product are more stable than moist one

- API: API also known as bulk active is the ingredient in a pharmaceutical drug that is biologically active and have the desired pharmacological effect on living beings. Ex: Amlodipine bisilate, Paracetamol
- Excipients: They are pharmacologically inert substances that are used for the proper delivery of drugs in their dosage form. Ex: Sorbitol , Aspartane

A brief classification of API on basis of therapeutic effect

1. 1.Anti-allergic drugs : These are pharmaceutical drugs or medicine, used to treat allergic reactions caused by foreign substances such as pollen, dust,mite. Ex: Cetirizine HCl
2. Antihistamine: Also known as histamine antagonist that are used for preventing and treating allergies, itching, hay fever etc. Ex: Loratadine
3. Antibiotics: These are are drugs used to stop or combat the growth of harmful microorganism like bacteria within the body without harming it. Also known as anti- effective drugs. Ex: Azythromycine
4. Anti-diabetics: Known as hypoglycemic agent, that are used for treating diabetics by lowering glucose level in blood. Ex: Insuline , Metformine
5. Analgesics: These are used to relive pain. Ex: Diclofenac Na, Paracetamol
6. Anti- neoplastic agent : Used for restricting or hindering of neoplasm (tumour) within the body. Ex: Bicalutamide
7. Anti cancer: These are used to kill cancer cell Within the body. Ex: carboplatine , Cisplatine
8. Anti viral: Used for treating curing and controlling viral infections. Ex: Acyclovir, Valacyclovir
9. Cardiovascular drug : These are given to patients suffering from heart and related disorder. Ex: Losartane potassium
10. Cough Medicine: Drugs that are used for treating and curing cough related syndromes. Ex: Cefixime
11. Anti fungal: Types of drug are crucial for treating, curing and preventing skin ailments. Ex: Fluconazole , Terconazole
12. Sergical Drug: Used to temporarily lower sensation or induce unconsciousness

Classification of Excipients

1. Anti adherent: Used to smoothen the surface of the tablet.
   i) Magnesium stearate
2. Binder: binders hold the ingredients in a tablet together.
   i) Manitol
3. Coating: i) Deterioration by moisture in the air
   ii) Unpleasant tasting tablets
   iii) Easier to swallow
   iv) Unwanted colour
   v) Odor masking
   vi) Physical and chemical protection
   vii) Protect the drug in the stomach. Eg. Synthetic polymers, shellac
4. Disintegrants: They ensure that when the tablet is in contact with water, it rapidly breaks down into smaller fragments, facilitating dissolution.
   Ex: Crosslinked polymers
5. Glidants: Glidants are substances that improve the flowability and dispersibility of powders by reducing friction. Can be inorganic materials such as talc and silica.
6. Filters and diluents: Used to increase the volume of small tablets. The drug become convenient for the consumer to use.
   Ex: i) Lactose
   ii) Sucrose
7. Flavours: To mask unpleasant tasting API. It may be synthetic or artificial. Non toxic
   Ex: i) Vanilla
8. Color:

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9. Lubricant: To smooth/polish the tablet surface. Also to increase the flow property in the machine.

i) Magnesium carbonate

10. Preservatives: For long durability of drugs.

i) Methyl paraben

11. Sorbent: Used for table/capsule moisture-proofing by limited fluid sorbing in a dry state.

Ex: TiO_2

12. Sweeteners: Ex: Sugar, Saccharine

- If all the test complies then the supplier is asked for sufficient amount sample to carry out the PD trial and QC test.
- PD will justify the production feasibility with supplied sample and QC will carry out 3 months study
- When the assessment of the questionnaire, PD trial report, Audit report are positive then the SCM negotiate the financial term with the supplier
- Provisionally approved
- Qualify the suppliers as approved vendor after receiving three consecutive lots with consistent quality.

Raw Material Procurement

- PD Product Development/ Research and Development design the new product and inform the SCM supply chain management
- SCM identify the sources of raw materials and send them vendor questionnaire
- Suppliers fill the questionnaire and return with sufficient amount of sample (100 gm) for both the API and excipients

  Also provide – Certificate of analysis (CoA)

  - Material safety data sheet(MSDS)
  - Stability data sheet (SDS)
  - Drug master file DMF
  - Good manufacturing practice (GMP)
- Quality Control QC The samples are received and send to the quality control department for analysis. QC carried out the full test according to the pharmacopeias and write down the comments.

Testing: A vast array of analyses on pharmaceutical raw materials (api+excipients) to ensure the quality and safety comparing standard specification.

According to GMP the specification must be written in precise terminology with detailed test method.

Sample/Raw Material
↓
Received
↓
Sampled
↓
Testing
↙ ↘
Accepted — Rejected
(Dispose)

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To ensure the quality following parameters are tested in QC Laboratory.

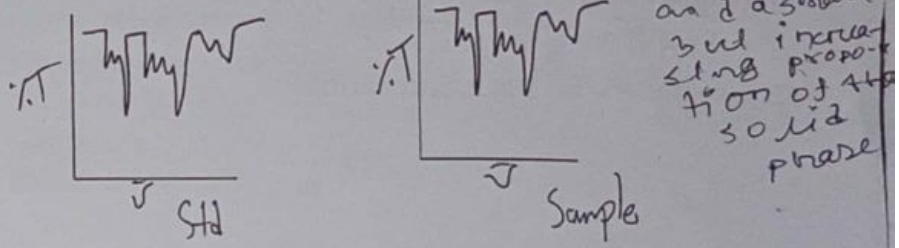
1. Physical and chemical test:
   a) Description: Colour/Odour/crystal/powder
   b) Solubility
   c) Identification : Compound to the standard material

All the properties should comply with the standard specification.

EPCRS (European pharmacopeia chemical reference substance)

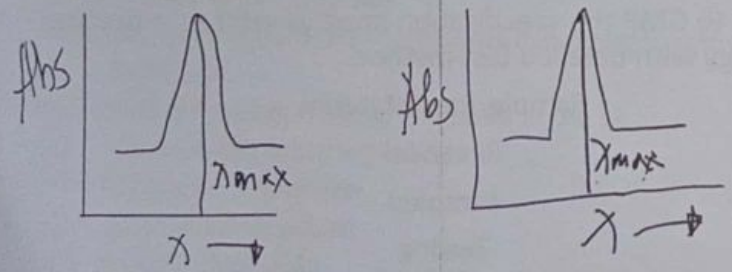
(C) Identification:
   i) Infra-red
   ii) UV
   iii) HPLC
   iv) Melting point
   v) Congealing point → is a point at which there exists a mixture of the liquid phase of a substance and a small but increasing proportion of the solid phase
   vi) TLC

IR



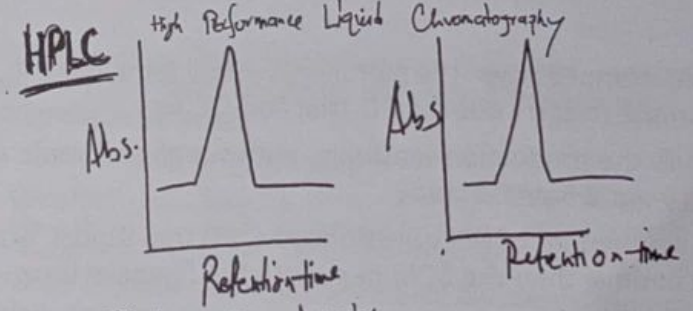
Write down two spectroscopic methods for identifying raw materials.

UV

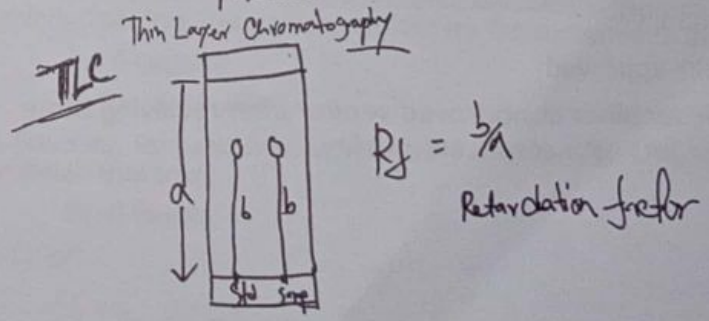


±2 nm acceptable
±2 nm acceptable

HPLC



TLC



$R_f = b/a$

Retardation factor

urity and quality:

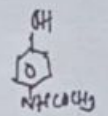
Level of impurity potency (% of the substance present in the sample)

1. General completeness of solution, pH, specific rotation, nonvolatile, residue on ignition loss on drying, water content, heavy metal. Ty volatile substance
2. Specific purity test: Ferric and Ferrous salt, peroxide, aldehyde, related disintegrated product.
3. Specific Quality test: Particle size, crystallinity, polymeric forms.
4. Assay. a) Calculated either on anhydrous or hydrous basis
   b) Total amount of sample (% unit)

2. Microbiological test: (I) Safety, potency, progenecity, sterility, anticeptic nature, toxicity

- Testing or assay of parcacetamol:

i) Description: White crystalline colourless powder,
Molecular formula > $C_8H_9NO_2$
Molecular Wt- 151.16 g/mol
Structural formula >



p-hydroxy acetanilide

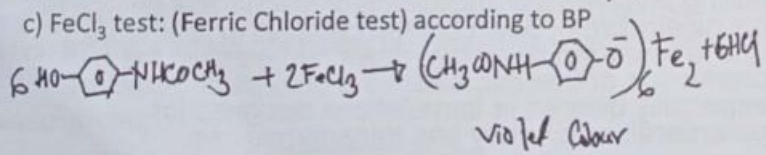
ii) Solubility: Sparingly soluble in water, freely soluble in alcohol, acetone, and very slightly soluble in DCM

3. Identification:

a) IR-Absorption spectrum, spectra superimposed to reference substance

b) UV-absorption spectrum – 249 nm = λmax

c) $FeCl_3$ test: (Ferric Chloride test) according to BP → British Pharmacopia.



Violet Colour

d) Melting point: 168-172 C°

4. pH range 5.3-6.5
5. Water content: Not more than 0.5%
6. Residue on ignition : Not more than 0.1%
7. Chloride: Not more than 0.014%
8. Sulphate: Not more than 0.02%
9. Sulfide: Absent
10. Heavy metals: Not more than 0.001%
11. Free amino phenol: Not more than 0.005%
12. D-chloroacetamide: Not more than 0.001%
13. LOD: Not more than 0.5%
14. Assay – 99-101%

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Pharmaceutical process

- API+Excipient
- Dispensing
- Mixing
- Milling
- Granulation
- Drying
- Blending
- Compression
- Coating
- Packaging

- Dispensing: Dispensing is the first and critical step in pharmaceutical manufacturing. During this step each ingredients are weighted according to dose. The following factors are important

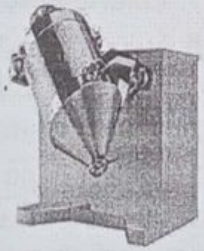


1. Wight accuracy
2. Dust control
3. HVAC (High vacuum air circulation)
4. Lot control
5. Material movement into and out of the dispensing should be considered

- MIXING : Mixing is a defined process that tends to results in randomization of dissimilar(size, colour, physical state differences) particles within a system. This is to be distinguished from order system in which the particles are arranged according to some interactive rules and thus follow a repetitive pattern.
- Two types of mixing

1. Fluid /liquid mixing
2. Solid mixing

Fluid + fluid
Solid + Solid



Quality of the instrument:

1. Physical properties of materials to be mixe
2. The economic consideration regarding the
3. Cost of equipment and maintenance
4. Quality of the materials to be used

- MILLING: Milling is the mechanical process of reducing the particle size of solids. By this process solid materials break into smaller pieces by grinding and crushing.

- Milling involves the application of mechanical energy to physically break down coarse particles to finer ones and is regarded as a "top–down" approach in the production of fine particles. Fine drug particulates are especially desired in formulations designed for parenteral, respiratory and transdermal use.

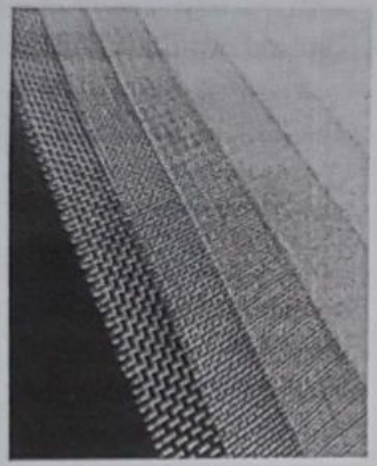


Milling equipment is classified as

1. Coarse: Larger than 20 mesh
2. Intermediate:200 to 20 mesh
3. Fine: Smaller than 200 mesh



Mesh: Number of opening per linear inch of screen



Pharmaceutical application of milling

1. In milling specific surface area increases by size reduction hence increase the therapeutic efficiency of medicinal compounds
2. The drying of wet masses are facilitated by milling.
3. The mixing and blending of several solid ingredients become easier if the ingredients are in same in size.
4. Milling of ointment, cream and paste provides a smooth texture and better appearance in addition to improve physical stability
5. Some factors should be considered in addition of standard adjustment of milling process.
   - Special atmosphere
   - Temperature control
   - Pre treatment
   - ual process milling compress milling
   - ubsequent treatment
   - Wet and dry milling

Drying

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- Drying applies for removal of a small amount of liquids from matters. Drying and evaporation are distinguished by the relative quantity of liquid removed from the solid.

Purpose of drying

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Expression of Drying

- Wet- weight basis: LOD (loss on drying): LOD is the loss of weight expressed as percentage w/w resulting from water and any kind of volatile matter that can be driven off under specific condition. LOD is an expression of moisture content in wet weight basis.

$$LOD(\%) = \frac{\text{wt. of water in sample}}{\text{total wt of wet sample}} \times 100$$

- Dry –weight basis: MC (Moisture content): the amount of water present in sample is calculated on dry weight basis

$$MC(\%) = \frac{\text{wt. of water in sample}}{\text{total wt of dry sample}} \times 100$$

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Difference of LOD and MC

- In LOD , sample is heated below its melting point in an oven and it includes all volatile matter including water content and solvents like alcohol

In MC only water is evaporated by Karl Fischer titration method.

- In LOD both bound and free water are evaporated

In MC free water is evaporated

- LOD values can vary from slightly above zero to slightly below 100. But MC value vary from slightly above zero to infinity

- Math: Wet weight of a sample is 4.5 g and dry weight is 2.5 g Find out the LOD and MC.

Wt of water in sample = (wet weight of sample – dry weight of sample)

=(4.5 -2.5) =2.0g

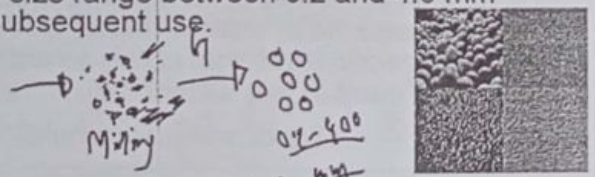
LOD = 2.0/4.5 * 100 = 44.44%

MC = 2.0/ 2.5 * 100 = 80%

Granulation

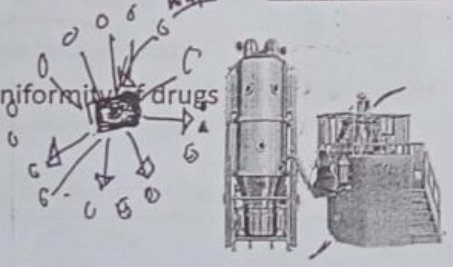
- Granulation is the process in which particles of raw materials are made to adhere to form larger granules in order to facilitate compression. granulation involves agglomeration of fine particles into larger granules, typically of size range between 0.2 and 4.0 mm depending on their subsequent use.

1. Dry granulation
2. Wet granulation



- Advantages

1. Ensure the higher distribution uniformity of drugs
2. Higher diffusion rate
3. Easy to compress
4. No introduction of heat
5. Continuous process



Blending

- Blending is a process of combining materials but blending is relatively technically gentle process compared to mixing . In this step lubricants and some excipients are added with the granules to ensure a homogenous mixture. Blending is a process that can be carried out numerous times within a manufacturing process when new excipients need to be added to the blend.
- Advantages:

1. Decreases the inherent cohesiveness and resistance to the movement between individual particles.
2. Make the surface of tablets smooth and polished.
3. Uniformity of granules
4. Increase the flow property of the granules
5. Used in the blending of lubricants , glidants or external disintegrants with granules



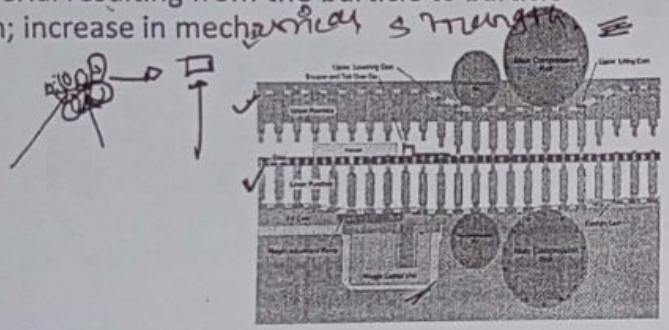
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Compression and Consolidation

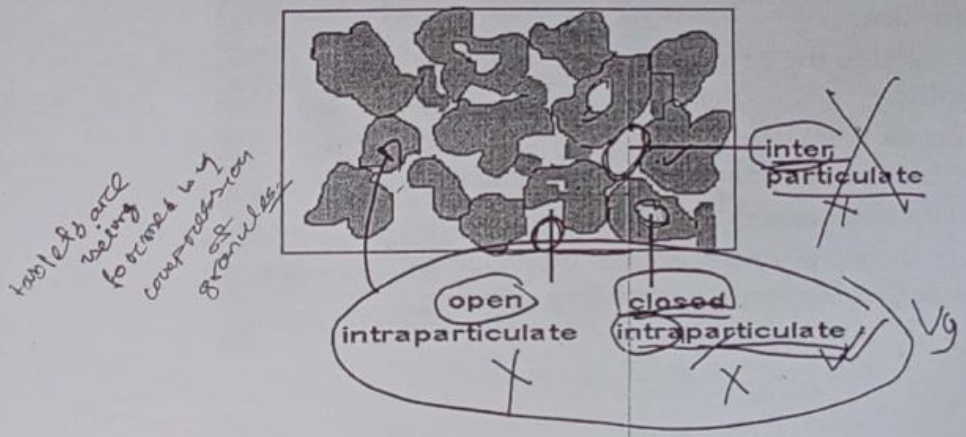
- Compression means the reduction of bulk volume of powdered solid as a result of displacement of gaseous phase by applied pressure.

Tablets are being formed by compression of granules.

Consolidation : It is an increase in the mechanical strength of the material resulting from the particle to particle interaction; increase in mechanical strength



Mass-Volume relationships



Three terms are used to define the volume of powdered solid / Mass volume relationship

- ***True Volume of the powder (V_t):*** True volume is the total volume of the solid particles. It is a volume of the particles excluding the inter and intra particulate spaces in a powder. Or it is volume of powder itself.
- ***Granular Volume of the powder (V_g):*** It is the cumulative volume occupied by the particles, including all intraparticulate (but not interparticulate) voids. Or it is the volume of powder itself and volume of intraparticulate spaces.
- ***Bulk Volume of the powder (V_b):*** It is the total volume occupied by the entire powder mass under the particular packing achieved during the measurement.

It comprises the true volume and inter and intra particulate voids.

- **Relative volume (V_R):** It is the ratio of the the volume, V of the sample under specific experimental conditions, to the true volume Vt.

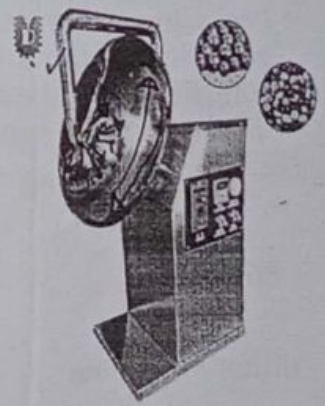
$V_R = V / V_t$

V_R tends to become unity as all air is eliminated from the mass during the compression process

Coating

Tablet coating is the last critical step in the tablet production. Coating is a process of spreading a specific solution of a substance over the surface of a tablet. Ex: propylene silicate, Opaglos 6000.

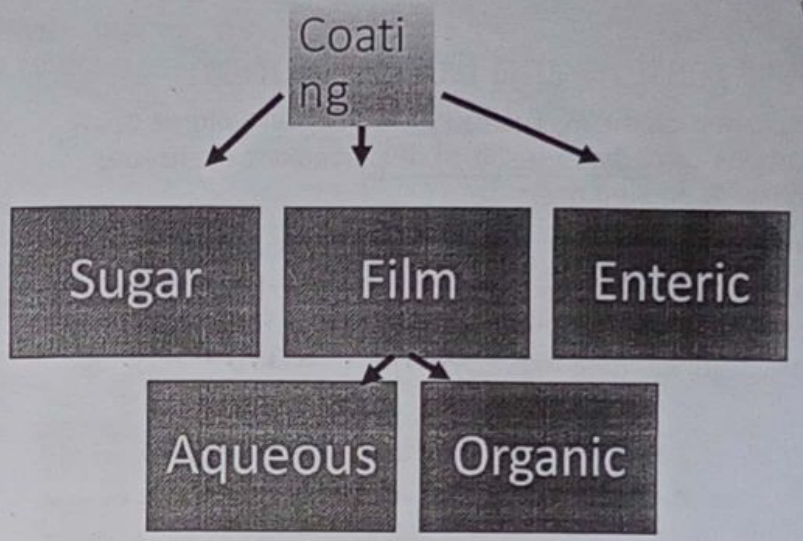
Tablet coating is a process by which an essentially dry, outer layer of **coating** material is applied to the surface of a dosage form in order to confer specific benefits over uncoated variety. **Coatings** may be applied to various oral dosage forms such as particles, powders, granules, crystals, pellets and **tablets.**



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OBJECTIVES OF TABLET COATING

- ✓ Mask the odour, taste or color of the drug.
- ✓ Provides physical and chemical protection for drug.
- ✓ Controls the release of drug from the tablet.
- ✓ Protects the drug from gastric environment of stomach in case of acid sensitive drug.
- ✓ Avoids chemical incompatibility.
- ✓ Improves pharmaceutical elegance by using colors and contrasting printing.



Packaging

- Packaging can be defined as an economic means of providing presentation, protection, identification, and compliance for the product during storage, carrying, display until the product is used for administration.

